

Aadarsha Gopala Reddy

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SUMMARY

M.S. Computer Science graduate (May 2026) from Washington University in St. Louis with hands-on experience across machine learning, data engineering, and full-stack development through research, teaching, and industry internships. Skilled in building end-to-end ML pipelines, real-time data systems, and production web applications with Python, PyTorch/TensorFlow, SQL, Spark/Kafka/Airflow, and modern JavaScript frameworks. Seeking software, data, or ML engineering roles.

SKILLS

Programming Languages: Java, C++, Python, R, HTML, CSS, JavaScript/TypeScript, PHP, C#, SQL, Rust.

Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, Node.js, Express.js, Vue.js, React, Socket.IO, Gemini/OpenAI API.

Tools & Technologies: Git, Tableau, Power BI, MySQL, AWS, MongoDB, Snowflake, Apache Airflow/Kafka/Spark, Streamlit, Microsoft 365, Google Workspace.

Languages: Proficient in English, Kannada, Telugu; Conversational in Hindi.

EDUCATION

M.S. in Computer Science

August 2024 - May 2026

Washington University in St. Louis, St. Louis, Missouri, USA

- *Thesis:* Toward Vehicle-Agnostic Driving Signatures for Cognitive Impairment Prediction from Naturalistic Driving Data
- *Projects:* Datacenter Cooling Optimization using Deep Reinforcement Learning; Multimodal Prediction of Alzheimer's Disease; MLB Statcast Real-Time Data Pipeline; Red-Blue Visual Auto Defender; Interactive Storybook; HackWashU Databases Workshop.
- *Leadership:* VP, Graduate-Professional Council (GPSC); Treasurer, HackWashU; Treasurer, Umang.

B.A. in Computer Science and Data Analytics; Economics Minor

August 2020 - May 2023

Ohio Wesleyan University, Delaware, Ohio, USA

- *Honors:* Golden Bishop Award, Mortar Board, Florence Leas Prize, Dean's List (Spring 2022, Spring 2023)
- *Projects:* Artificial Intelligence in Modern Board Games; Connect 4 AI; Artificial Intelligence Opinion Survey; AIParkinScan.
- *Leadership:* VP of Finance (Campus Programming Board); Budget Committee Senator (WCSA); Treasurer & Founding Member (Hindu Student Council).

EXPERIENCE

Graduate Teaching Assistant - CSE 5114: Data Manipulation and Management at Scale

January 2026 - May 2026

Washington University in St. Louis, St. Louis, Missouri, USA

- Taught large-scale data pipeline concepts (Snowflake, Airflow, Spark, Kafka, Flink) to a graduate cohort, covering fault-tolerant design, streaming windowing strategies, and query optimization.
- Designed oral midterms as mock technical interviews, requiring students to reason through pipeline architecture decisions and troubleshoot system failures under real-world constraints.
- Mentored three capstone teams from raw open datasets to production-grade dashboards, providing weekly feedback that kept all teams on schedule through final presentations.
- Held weekly office hours to debug student pipeline implementations across Snowflake schema design, Airflow DAG failures, and Kafka consumer configuration.

Master's Thesis Research

August 2025 - May 2026

DRIVES Project, Washington University in St. Louis, St. Louis, Missouri, USA

- Built an end-to-end pipeline processing 26,968 participant-weeks of naturalistic driving data from 304 older adults, engineering vehicle-agnostic weekly behavioral features and integrating demographic covariates for CDR status prediction.
- Benchmarked six prediction models (GRU-DANN, DANN, logistic regression, random forest, XGBoost, MLP) under leave-one-group-out cross-validation; GRU-DANN achieved the highest participant-level ROC AUC (0.599) and balanced accuracy (0.584).

AI Engineer Intern

June 2025 - August 2025

Crittero, Inc., Remote

- Generated labeled social media training data for seven user personas via time-series behavioral modeling in TypeScript, capturing post sequences, engagement timings, and content preferences.
- Trained a TensorFlow/Keras recommendation model with persona-type, content-category, and temporal embeddings, plus a collaborative-filtering cold-start fallback for new users.
- Built a configurable Python CLI for reproducible model evaluation, enabling systematic comparison across hyperparameter settings, embedding dimensions, and fallback thresholds.

Graduate Assistant with the Taylor Family Center for Student Success

August 2024 - May 2026

Washington University in St. Louis, St. Louis, Missouri, USA

- Planned and ran three recurring engagement programs for first-generation, limited-income (FLI) students — Welcome Dinner, First-Gen Week, and End of Week Unwind — coordinating logistics, promotion, and on-the-day operations.
- Authored FERPA-compliant process manuals for engagement data collection and evaluation, giving center staff a standardized, repeatable framework for measuring FLI student participation.
- Analyzed semester survey responses and event attendance data, producing reports that directly informed the following cycle's programming decisions and outreach priorities.

Data Analyst Intern

June 2023 - May 2024

Lab714, Boca Raton, Florida, USA

- Analyzed IoT environmental sensor data on AWS, building client-facing dashboards that surfaced chemical imbalances and equipment alerts — helping clients cut pool maintenance costs by 20%.
- Built AWS data ingestion pipelines to normalize and store continuous sensor telemetry from deployed devices, enabling live dashboard readings for client operational decisions.
- Wrote SQL queries to aggregate historical sensor data by pool, date range, and chemical parameter, powering the trend analytics and anomaly detection shown in client dashboards.
- Designed and engineered the Figma-to-React UI for an all-in-one environmental sensor device, delivering a pixel-perfect, production-ready interface from initial wireframes.

Computer Science Lab Assistant

January 2022 - May 2022

Ohio Wesleyan University, Delaware, Ohio, USA

- Tutored 10+ students weekly across introductory through advanced CS courses, reinforcing concepts in algorithms, data structures, and programming fundamentals.
- Graded assignments for roughly 35 students in introductory and intermediate courses, providing detailed feedback to help students improve coding and problem-solving skills.

Media Center Assistant

September 2021 - May 2022

Ohio Wesleyan University, Delaware, Ohio, USA

- Set up and tore down audio-visual equipment for events across campus, providing on-site troubleshooting to ensure smooth operation for organizers and attendees.
- Managed room reservations and tracked equipment loans using Sierra, coordinating with event organizers to fulfill audio-visual needs across campus venues.